English Writing

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Abstract

This is my academic writing note. Try to use this note to improve my writing skills.

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Chapter 1

The general principle of academic writing

1.1 Some of my weakness

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Examples
How to improve my writing skills?

Chapter 2

The structure of the TACAS paper

2.1 Title

2.2 Abstract

- Explicitly point out soundness issues happened in GPU and MILP.
- Use continuous constraint programming (abstract satisfaction) to examine neural network verification.
- Mention 3 different propagators and the underlying parallel framework of SymJET is lock-free asynchronous.
- All the results are machine-checked by autoformalization.
- The comparison of experimental results of different solvers.

2.3 Introduction

- Simply introduce neural network verification.
- There are bunch of neural network verifiers, but these tools typically have soundness issues:
  - Ignore FP rounding errors on GPU.
  - Use MIP solvers.

  These results are demonstrated in the experimental section.
- How to reduce the soundness issue? What is the position of this paper?
  - Use CCP.
  - The connection between NNV and CCP.
  - How to formalize NNV by CCP?
  - The proposed CCP solver is lock-free.
- Contributions
- Introduce autoformalization, and how it is going to be used through the paper.

2.4 Abstract Satisfaction

- Introduction
- Logical concrete domain
- Floating-point interval abstraction

- Galois connection
- Soundness and interval propagators
- Parallel greatest fixpoint

2.5 NNV by Constraint Propagation

- List out our contributions
- Neural concrete domain
- Neural network verification problem
- Feedforward neural networks
- From neural to constraint representation

2.6 Specialized Neural Network Interval Abstraction

- Mention the connection with the previous section, and contributions.
- Graph-based propagator
- Symbolic propagation

2.7 Experimental Results

- Environment settings
- The soundness issue on GPU
- VNN-COMP 2025
- Discussion (removed)

2.8 Conclusion

Chapter 3

The structure of the CPAIOR paper